

NETWORKWALKS ACADEMY

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

Project Information	Details
Report week	Week 02
Modules completed	Module 1 – Footprinting & Reconnaissance Module 5 – Network Scanning with Zenmap
Pentester name	Lakkappa Padmanna Pujer
Program / Batch	B082 – Networkwalks
Target 1	networkwalks.com – authorized training target
Target 2	10.221.30.0/24 – local lab network
Platforms	Kali Linux and Windows PC with Zenmap
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery

Liability Disclaimer

All activities documented in this report were performed for the assigned educational cybersecurity project and within the stated authorized scope. Reconnaissance and scanning should only be conducted against systems and networks for which appropriate permission has been obtained. The findings below are observations from the practical exercises and are not, by themselves, proof of exploitable vulnerabilities.

Prepared from the supplied Networkwalks task material, sample report format, and execution screenshots.

1. Introduction

This report follows the structure of the supplied Networkwalks sample report: project information, liability disclaimer, introduction, tools used, activities performed, risk/impact, recommendations, conclusion and evidence collected. The sample report covers W2-PM1 and W2-PM5 and states that each activity should include the command, observed result, screenshot evidence and a short explanation. [filecite2turn1file0L2-L18](#)

For this submission, the requested report label is Week 01. The supplied sample document itself labels the underlying modules as W2-PM1 and W2-PM5 / Week 2; the report therefore keeps the module names while using Week 01 as the requested submission label. [filecite2turn1file0L8-L18](#)

2. Tools Used

Tool / Platform	Purpose
Kali Linux	Operating system used for reconnaissance activities
WHOIS	Domain registration details and name servers
WhatWeb	Web technology fingerprinting
nslookup	DNS name resolution
curl -I	HTTP response-header inspection
wafw00f	Web Application Firewall detection
dnsrecon	DNS record enumeration
Windows CMD / ipconfig	Local IP address, subnet and gateway
Zenmap (Nmap GUI)	Live-host discovery, service scanning and topology visualization

3. Activities Performed

3.1 Module 1 — Footprinting & Reconnaissance

The supplied practical requires six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon. [filecite2turn0file2L17-L22](#)

Task 1 — WHOIS Domain Registration Lookup

Command: whois networkwalks.com

The supplied evidence shows the domain registration record, including GoDaddy.com, LLC as registrar, creation date 2019-11-06, expiry date 2027-11-06, and HostGator name servers.

Field	Observed result
Registrar	GoDaddy.com, LLC
Creation date	2019-11-06
Expiry date	2027-11-06
Name servers	NS6135.HOSTGATOR.COM / NS6136.HOSTGATOR.COM
DNSSEC	Unsigned

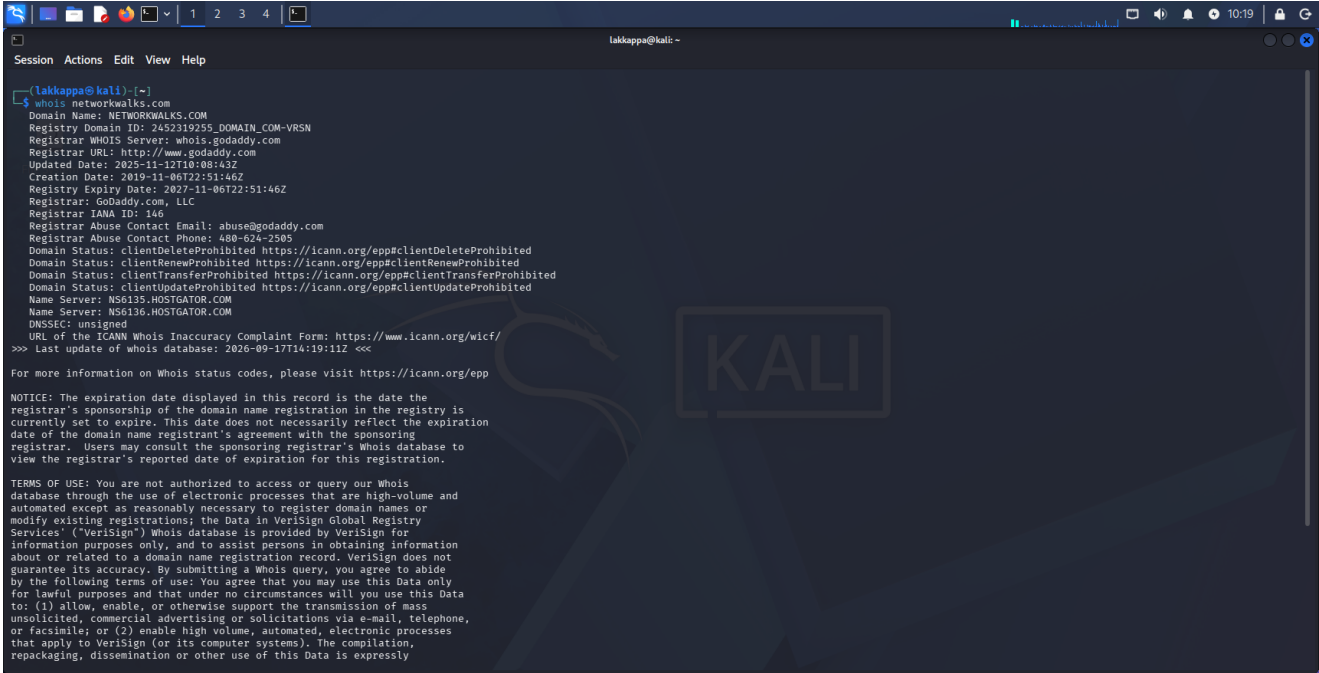


Figure 1. WHOIS command evidence.

Task 2 — Web Technology Fingerprinting

Command: whatweb networkwalks.com

The supplied WhatWeb evidence identifies Apache, WordPress, WP Download Manager and jQuery, together with the public IP and other web metadata.

Observed item	Result
Web server	Apache
CMS	WordPress 7.1 (as shown in supplied screenshot)
Plugin	WordPress Download Manager 3.3.58
JavaScript library	jQuery 3.7.1
Public IP	192.232.216.135

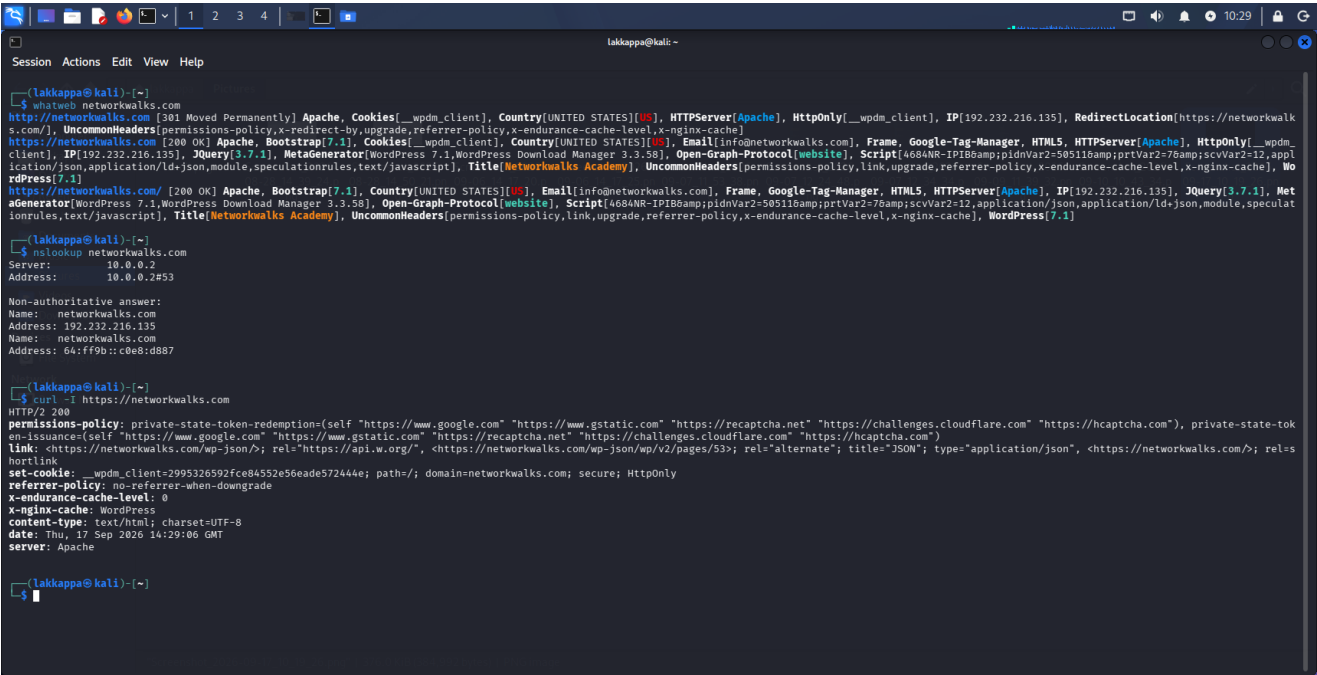


Figure 2. WhatWeb, nslookup and curl evidence.

Task 3 — DNS Resolution

Command: nslookup networkwalks.com
 The supplied output resolves networkwalks.com to 192.232.216.135. An IPv6-formatted result is also displayed.

DNS item	Observed result
IPv4	192.232.216.135
IPv6	64:ff9b::c0e8:d887
DNS server shown	10.0.0.2#53

Task 4 — HTTP Response Header Inspection

Command: curl -I https://networkwalks.com
 The supplied output reports HTTP/2 200, Apache, WordPress-related headers, the /wp-json/ REST API link and the __wpdm_client cookie. The sample report similarly explains that Curl was used to inspect HTTP response headers and the WordPress REST API endpoint. [filecite?turn1file0?L60-L69](#)

Task 5 — WAF Detection

Command: wafw00f networkwalks.com
 The supplied practical evidence reports that the website is behind ModSecurity (SpiderLabs).

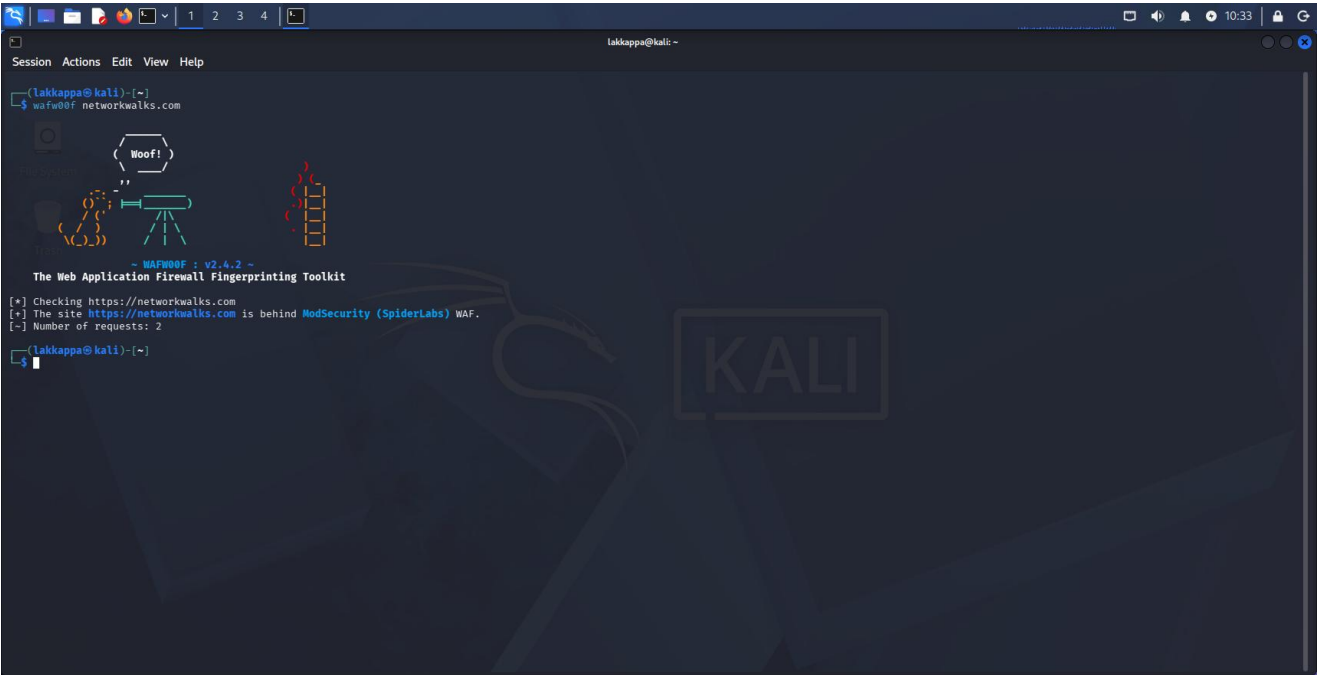


Figure 3. WAF fingerprinting evidence showing ModSecurity (SpiderLabs).

Task 6 — DNS Record Enumeration

Command: `dnsrecon -d networkwalks.com`

The supplied DNSRecon output reports SOA, NS, MX, TXT and SRV records and a total of 16 records.

Record type	Observed result
SOA / NS	ns6135.hostgator.com and ns6136.hostgator.com
DNS software	BIND 9.16.23-RH
MX	mail.networkwalks.com → 192.232.216.135
TXT / SPF	v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
SRV	cPanel autodiscover records to cpanelmaildiscovery.cpanel.net
Records reported	16

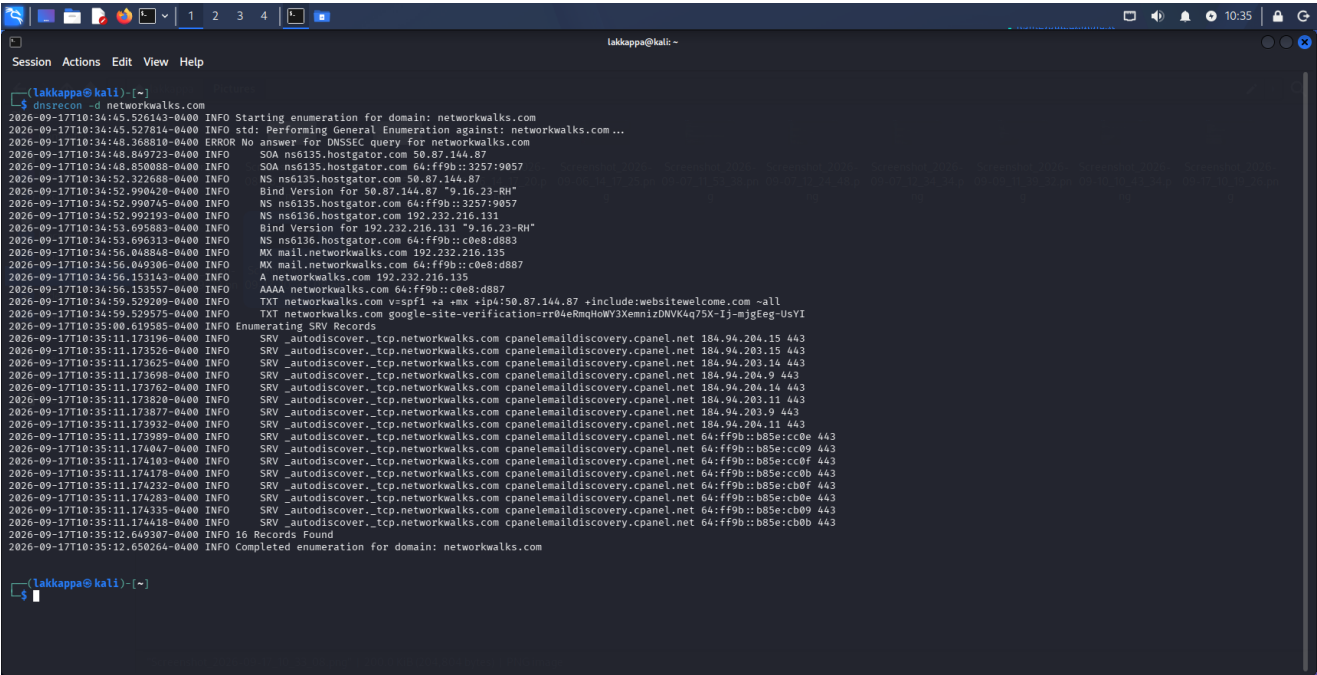


Figure 4. DNSRecon enumeration evidence.

3.2 Module 5 — Network Scanning with Zenmap

The supplied PM5 practical requires identifying the local IP/subnet, discovering live hosts, identifying IP/MAC addresses and generating a network topology. [filecite?turnOfile1?L18-L28](#)

Task 1 & 2 — Zenmap Setup and Local Network Identification

The Windows ipconfig evidence shows the active Wi-Fi adapter with IPv4 address 10.221.30.116, subnet mask 255.255.255.0 and default gateway 10.221.30.123. The corresponding /24 subnet is 10.221.30.0/24.

Parameter	Observed value
IPv4 address	10.221.30.116
Subnet mask	255.255.255.0
Derived subnet	10.221.30.0/24
Default gateway	10.221.30.123

```

Command Prompt
Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : fe80::4e31:a86b:63d7:e0ab%22
IPv4 Address. . . . . : 192.168.52.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 

Ethernet adapter VMware Network Adapter VMnet8:
Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : fe80::3811:ad64:cdf4:8161%3
IPv4 Address. . . . . : 10.0.0.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 

Ethernet adapter VMware Network Adapter VMnet2:
Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : fe80::650:58b8:2879:7b6c%19
IPv4 Address. . . . . : 192.168.75.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 

Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . : 
IPv6 Address. . . . . : 2409:40f2:101a:356c:3ad4:7175:c087:806a
Temporary IPv6 Address. . . . . : 2409:40f2:101a:356c:7033:d43b:c263:4127
Link-local IPv6 Address . . . . . : fe80::a706:f98d:63b6:7aac%13
IPv4 Address. . . . . : 10.221.30.116
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::b004:72ff:fe79:ff5f%13
                             10.221.30.123

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : shrideviengineering.org

C:\Users\lakka>
  
```

Figure 5. Windows ipconfig evidence.

Task 3 — Discover Live Hosts

Zenmap was configured for 10.221.30.0/24 with the Ping Scan profile. The generated command was `nmap -sn 10.221.30.0/24`. The supplied scan output reports 256 addresses scanned and one host up: 10.221.30.116.

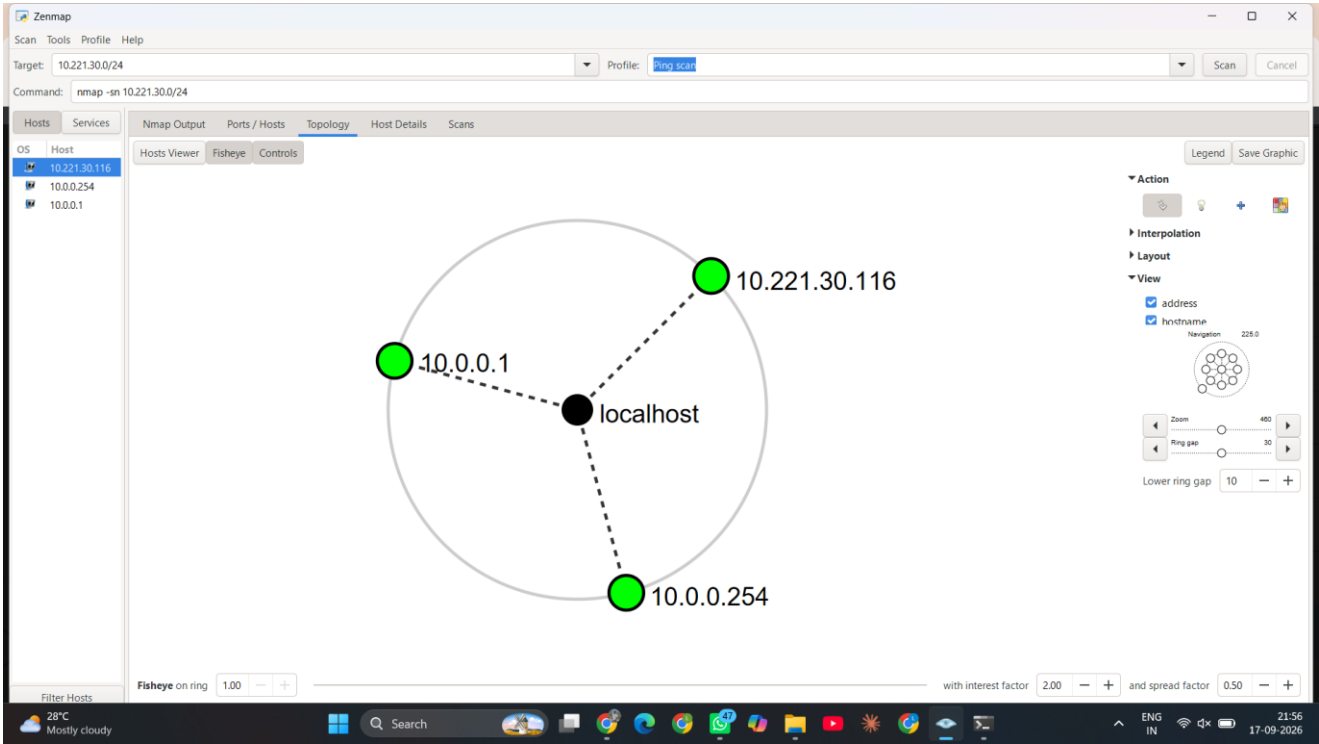


Figure 6. Zenmap Ping Scan evidence for 10.221.30.0/24.

Task 4 & 5 — Number and IP Address of Live Hosts

Item	Observed result
Addresses scanned	256
Hosts up	1
Live host	10.221.30.116
Reported scan time	12.20 seconds

Task 6 — MAC Address

The supplied Ping Scan screenshot does not visibly show a MAC address for 10.221.30.116. Therefore, this report does not invent a MAC value. The PM5 task sheet requires MAC identification and notes that results vary according to the actual network configuration. 2filecite2turn0file12L51-L692

Supplementary Zenmap Intense Scan Evidence

The supplied screenshots also contain an additional Intense Scan against 10.221.30.0/24. This is retained as supplementary evidence, separate from the core PM5 task list. The output for 10.221.30.116 shows open TCP services including 135, 139, 445, 903, 3306 and 5432.

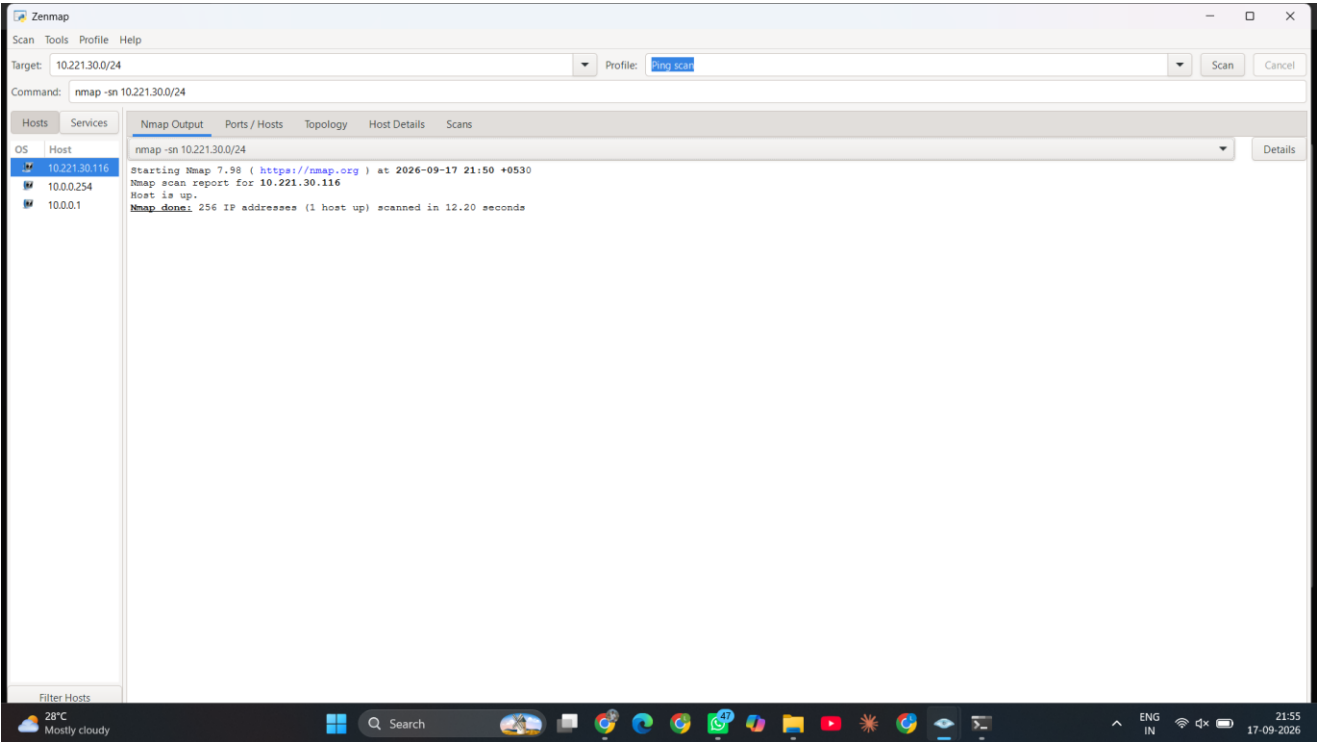


Figure 7. Supplementary Zenmap Intense Scan evidence.

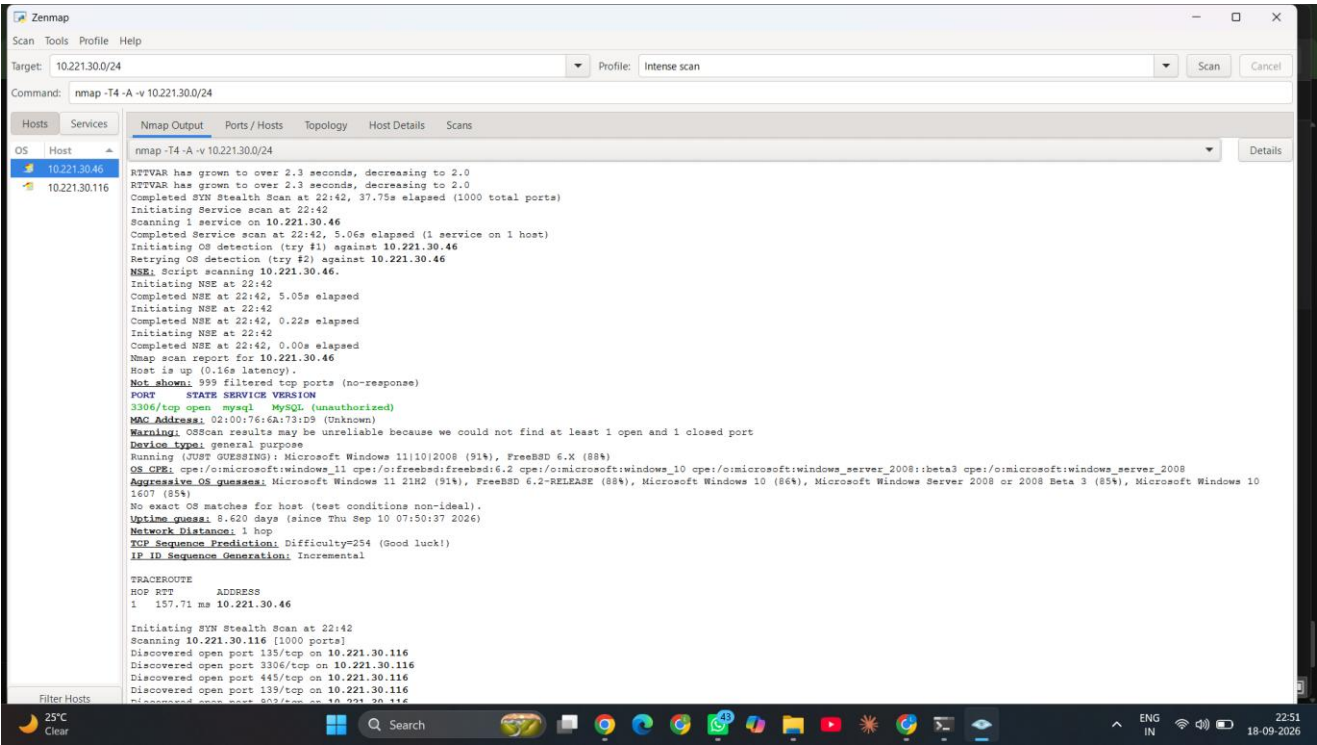


Figure 8. Supplementary service/OS detection evidence for 10.221.30.116.

Task 7 — Zenmap Topology and PDF Export

The supplied topology screenshot shows localhost connected to 10.221.30.116 and also displays 10.0.0.1 and 10.0.0.254 retained in the Zenmap session. The PM5 task requires opening Topology, enabling the legend and saving the topology as PDF. [filecite2turn0file1L64-L74](#)

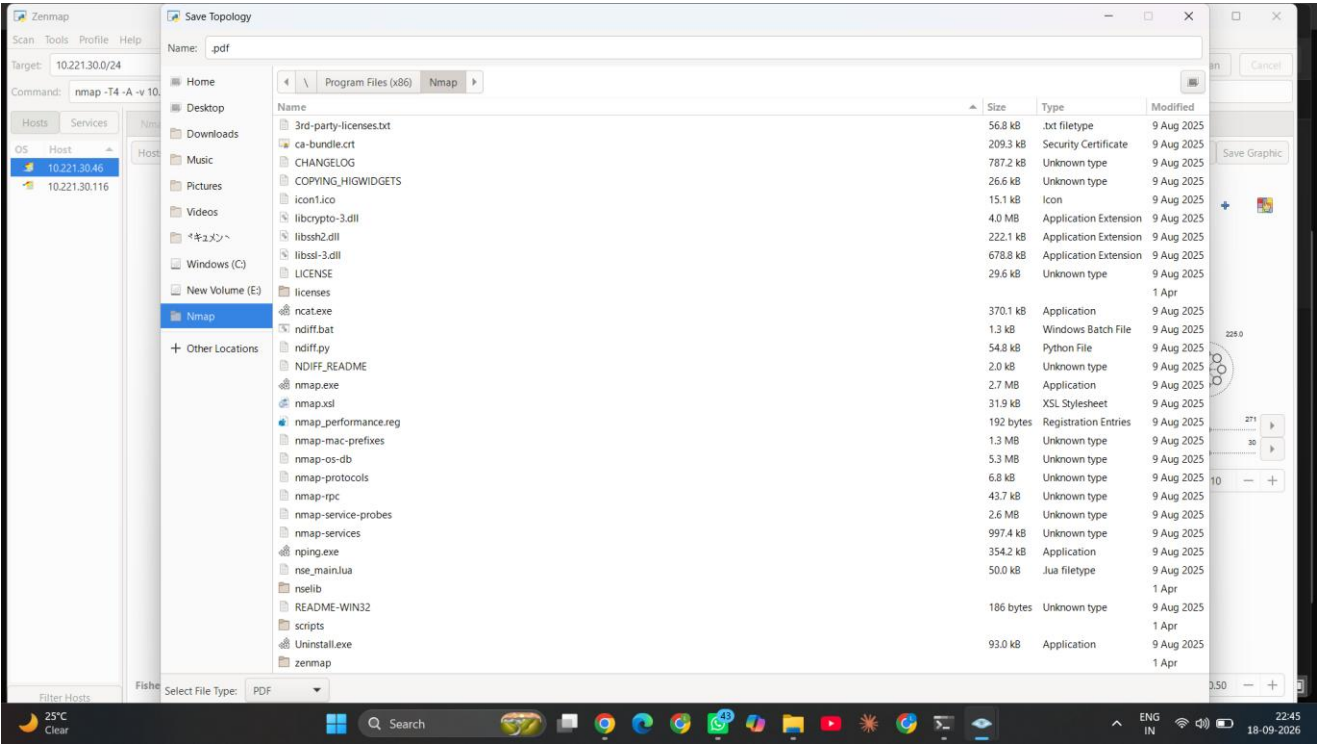


Figure 9. Zenmap Topology evidence.

4. Risk Analysis / Impact

The sample report presents findings as potential risks rather than confirmed vulnerabilities and explicitly states that information such as software versions, IP addresses and DNS records does not by itself prove a vulnerability.

#	Finding / observation	Potential impact	Level
1	Web technology information exposed	May assist technology fingerprinting and further security review	Medium
2	Server IP identifiable	Provides information about the network location of the web service	Low
3	HTTP technical information exposed	May assist fingerprinting and further enumeration	Low
4	WAF technology identifiable	Reveals information about security architecture	Low
5	DNS infrastructure information exposed	Can help build a broader infrastructure profile	Medium
6	Live host visible on local network	Unexpected devices may require investigation	Medium

These are observations from reconnaissance and scanning, not confirmed vulnerabilities. No exploitation or vulnerability validation is documented for these two modules.

5. Recommendations

- Review publicly exposed technology information** — Regularly review what web technologies, CMS components and plugins are publicly visible.
- Keep software updated** — Regularly update CMS platforms, plugins and other web technologies and review current security advisories.
- Review HTTP headers** — Check response headers for unnecessary technical information.
- Review DNS records regularly** — Ensure that only required information and services are publicly exposed.
- Properly configure and monitor the WAF** — Keep ModSecurity enabled and appropriately tuned.
- Perform regular internal network discovery** — Periodically scan owned/authorized networks to identify active devices.
- Investigate unknown devices** — Verify any unexpected device discovered during network scanning.
- Maintain network documentation** — Keep topology and device information current.

9. Perform security testing with authorization — Conduct reconnaissance and scanning only within approved scope.

6. Conclusion

During the requested Week 01 project submission, I documented practical activities covering footprinting/reconnaissance and network scanning. Module 1 used six Kali Linux tools to collect domain, web, DNS, HTTP-header and WAF information. Module 5 used Windows ipconfig and Zenmap to identify the local 10.221.30.0/24 subnet and discover 10.221.30.116 as the live host in the supplied Ping Scan.

The exercises demonstrate how structured information gathering can be followed by controlled network discovery. The report also separates direct observations from supplementary scan output and does not treat the observed information as proof of a vulnerability.

The sample report emphasizes clear evidence, documented findings and authorized scope; those principles are followed here.

7. Evidences Collected

The following figures are the execution screenshots supplied for this project and embedded above in the relevant task sections.

Figure	Evidence	Purpose
1	WHOIS	Domain registration and name servers
2	WhatWeb / nslookup / curl	Web stack, DNS resolution and HTTP headers
3	Wafw00f	ModSecurity (SpiderLabs) detection
4	DNSRecon	DNS, mail, TXT and SRV enumeration
5	Windows ipconfig	Local IP, subnet and gateway
6	Zenmap Ping Scan	Live-host discovery
7	Zenmap Intense Scan	Supplementary scan evidence
8	Zenmap service/OS output	Supplementary open-service evidence
9	Zenmap Topology	Topology visualization / PDF-export evidence

- End -

Author: Lakkappa Padmanna Pujer | Program: Cybersecurity & Ethical Hacking | Networkwalks